COMETFWC067
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LED Interfacing with ESP32-WROOM Board

Implementing a Simplified Boolean Expression through GPIO Output

Aim

To interface an LED with the ESP32-WROOM development board by implementing the simplified Boolean expression f = P xor Q xor R through its GPIO output, and to verify on hardware that the LED turns ON for logic HIGH and remains OFF for logic LOW, in agreement with the derived truth table.

GATE Question

The Boolean expression for the output f of a 4:1 multiplexer with inputs I0 = R, I1 = R', I2 = R', I3 = R and select lines S1 = P, S0 = Q is one of the following:

Option	Expression
A	(P xor Q xor R)'
B	P xor Q xor R
C	P + Q + R
D	(P + Q + R)'

Question Analysis

1. Select-Line to Output Mapping

S1 (P)	S0 (Q)	Line Selected	Output f
0	0	I0	R
0	1	I1	R'
1	0	I2	R'
1	1	I3	R

2. Sum of Products Form

f = P'Q'R + P'QR' + PQ'R' + PQR

3. Observation and Final Expression

f is 1 only where P, Q, R contain an odd number of 1's (001, 010, 100, 111) - the defining property of the three-input XOR operation.

Therefore, f = P xor Q xor R, which matches **Option (B)**.

Correct Answer: (B) f = P xor Q xor R

Components Used

S.No.	Component
1	ESP32-WROOM Development Board
2	USB Cable
3	Breadboard
4	LED
5	Jumper Wires

Purpose of Each Component

Component	Function
ESP32-WROOM Development Board	Runs the flashed firmware and generates the GPIO output corresponding to the simp
USB Cable	Connects the ESP32 board to the computer for flashing the firmware and powering t
Breadboard	Provides a platform to build and hold the LED circuit
LED	Displays the resulting output logic level - ON for HIGH, OFF for LOW
Jumper Wires	Connects the ESP32 GPIO pin to the LED circuit on the breadboard

Hardware Connections

S.No.	ESP32-WROOM Pin	Connected To
1	GPIO 2	LED Anode (+)
2	LED Cathode (-)	GND
3	USB Port	Computer (for programming)

Execution Procedure

1. A PlatformIO project was set up for the ESP32-WROOM board (board type esp32dev).

2. The source code implementing the Boolean expression f = P xor Q xor R on GPIO 2 was placed inside the src folder of the project.

3. The project was built to generate the firmware, bootloader, and partition binary files (bootloader.bin, partitions.bin, firmware.bin).

4. The ESP32 board was connected to the Ubuntu PC through a USB cable, and the system was checked to confirm the board was detected as a serial device (/dev/ttyUSB0).

5. The chip identity was verified using the esptool.py chip_id command.

6. The generated binary files were flashed onto the ESP32 at their respective memory offsets using esptool.py write_flash.

7. Once flashing completed, the EN (reset) button on the board was pressed to run the new firmware.

8. The LED connected to GPIO 2 was observed to confirm that its ON/OFF behaviour matched the expected logic output of the Boolean expression.

Truth Table

P	Q	R	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Results and Observations

The Boolean expression f = P xor Q xor R was successfully implemented on the ESP32-WROOM development board. After flashing the generated binaries using esptool.py, the LED connected to GPIO 2 turned ON whenever the output f was logic HIGH and remained OFF whenever f was logic LOW, exactly matching the truth table derived from the GATE question. This confirmed that the 4:1 multiplexer expression simplifying to the three-input XOR function was correctly realized in hardware using the ESP32 board.